BURNBENCH · RESEARCH REPORT · AUGUST 2026

The Stack Is the Model

Which agent harness × model actually ships. All 40 stacks: 5 frontier models crossed with 8 agent harnesses over 33 verified software engineering tasks, with hidden-oracle verification and no self-reported outcomes.

MODELS

HARNESSES

5,233 VERIFIED RUNS 1,247 model × harness × task cells	40 STACKS MEASURED 5 models × 8 harnesses, crossed	33 TASKS hidden oracle tests, 3-5 trials
51.2% VARIANCE FROM HARNESS vs 12.4% from the model	-0.95 FLAKE × QUALITY determinism predicts pass rate	126× COST PER PASS SPREAD same tasks, same outcomes

VERIFICATION STATUS

Reviewed by two independent adversarial audits. Both reproduced the arithmetic from source and both returned HOLD on publication. Every finding they raised is applied or disclosed in this document; the corrections register is on page 8. Figures that moved under audit are reported as robustness bands across specifications rather than as point estimates.

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EXECUTIVE SUMMARY

Six findings, with their robustness

Each finding below is stated with the specification it depends on. Where an adversarial audit showed a number moves under a different but defensible specification, the range is printed rather than the most flattering endpoint. Findings marked ROBUST hold under every specification tested.

01 The harness explains more than the model

ROBUST

Harness 51.2% · model 12.4% · interaction 36.3%

Two-way variance decomposition over 40 stacks. On the balanced 25-task set common to all stacks it is harness 54.1% / model 15.5% / interaction 30.3%. The ordering survives every specification.

02 Harness rankings transfer poorly between models

QUALIFIED

Mean Kendall τ = +0.286 to +0.400

Unbalanced +0.286; per-pair balanced +0.336; global common-task set +0.400. Direction holds throughout, magnitude does not. The honest reading is weak-to-moderate agreement, not the near-zero the unbalanced figure suggests.

03 One harness is bad everywhere; none is good everywhere

ROBUST

OpenCode ranks last for 4 of 5 models, seventh for the fifth

Mean strict rate 61.7%, 20 points below the next harness, with a rank range of 1 across models. No harness leads more than three of the five models, and every other harness swings 2 to 6 places depending on the model it is paired with.

04 Determinism is the strongest correlate of quality

QUALIFIED

r = -0.950 between flake rate and pass rate

308 of 1230 multi-trial cells (25.0%) both passed and failed the same task. Excluding the OpenCode outlier the correlation is -0.839 over 7 points. Both are far above tokens (+0.315), cost (+0.219) and latency (+0.115).

05 Cost is close to orthogonal to capability

ROBUST

126× spread in cost per passing task

$0.0028 to $0.3453 for the same task set. 12 domination pairs across 5 distinct configurations, where a harness at least 3× cheaper on the same model also passes more.

06 A saturated suite does not predict hard-task performance

ROBUST

Main-suite vs hard-band rank: τ = +0.149

Over 32 stacks. Between-stack SD rises from 0.086 on saturated tasks to 0.258 on the hard band. Verified robust across 2,000 random tie-breaks by external audit.

SECTION 01 — DESIGN AND COVERAGE

What was actually run

Five models, eight harnesses, 33 tasks with measured cells out of 55 defined, three to five trials per cell, all inside the same container image against pinned model endpoints. Pass and fail are decided by oracle test suites held outside the workspace and executed after the agent exits. No self-report, no diff inspection, no judge in the pass/fail loop.

COVERAGE IS NOT COMPLETE, AND THIS MATTERS

Only 25 of 33 tasks were run by all 40 stacks. An earlier export of this data let DeepSeek's Prime Agent run 13 easy tasks its seven peer harnesses never saw, inflating its DeepSeek rank to 2; that composition artifact was repaired before publication and Prime Agent now ranks 7 of 8 on both the published and balanced views. Every cross-model statistic in this report is still reported both as published and as recomputed on balanced task sets.

Task coverage by stack

Model	Tasks per harness	Balanced set	Runs	Note
DeepSeek V4	32–32	32	1,186	uniform across harnesses after repair
GLM 5.2	31–33	30	1,113	full band incl. 33-37
GPT 5.6 Luna	32–32	32	1,155	full band incl. 33-37
Muse Spark 1.2	32–33	31	1,134	full band incl. 33-37
Qwen3.7 Flash	26–28	26	645	hard band 33-37 not run

The 25-task globally common set spans 01-02, 05-10, 13-26 and 30-32. It excludes the hard band (33-37), which Qwen never ran, and tasks 03-04 and 11-12, whose verifiers failed or were repaired under audit. Statistics computed on it are balanced but narrower in scope; statistics computed on everything are broader but composition-sensitive. Both are given throughout.

Outcome definitions

Metric	Definition
strict_rate	Fraction of trials where the hidden oracle suite passes cleanly. The headline metric.
solution_rate	Strict passes plus trials the verifier accepts under relaxed timing. Global gap: 2.9 points.
quality_mean	Kimi judge, rubric v1, 1-5, scored on the untruncated diff. Never influences pass/fail.
non-deterministic	A multi-trial cell where 0 < strict_passes < n. The same configuration both passed and failed.
cost_usd_median	Median billed cost per task, derived from token counts at endpoint list price.

SECTION 02 — FINDING 01

The harness explains more than the model

Every agent evaluation in circulation ranks models. We ran the cross product and decomposed where the variance actually lives. Under both the published and the balanced specification, the harness main effect exceeds the model main effect.

Specification	Stacks	Model	Harness	Interaction
All tasks, 8 harnesses	40	12.4%	51.2%	36.3%
Balanced 25-task set, 8 harnesses	40	15.5%	54.1%	30.3%
All tasks, excluding OpenCode	35	29.5%	12.9%	57.6%
Balanced set, excluding OpenCode	35	31.3%	23.3%	45.4%

WHERE THIS CLAIM STANDS AFTER REPAIR

Earlier drafts, audited against the unrepaired export, reported that removing OpenCode inverts the decomposition on the balanced set (model 44.1%, interaction 23.8%). That inversion does not reproduce on the corrected export. With the coverage defect repaired, the interaction term remains the largest share in every specification — 57.6% unbalanced, 45.4% on the balanced 25-task set — with the model second at 31.3%. The pairing, not either component alone, is where most of the variance lives.

Magnitudes matter as much as shares. Swapping harness while holding the model fixed moves the pass rate by a median of 4.9 points and a mean of 10.1 points across all 140 within-model harness pairs. Swapping model while holding harness fixed moves it a median of 5.5 and a mean of 8.5 points. The best-to-worst range within a model averages 31.1 points, which is a range, not a typical swap, and earlier drafts conflated the two.

The full cross product

Model	Claude Code	Codex	Droid	Hermes	OMP	OpenCode	Pi	Prime Agent
DeepSeek V4	82.0%	**90.0%**	84.0%	82.7%	85.3%	72.0%	82.9%	80.1%
GLM 5.2	88.7%	85.3%	89.2%	88.3%	**90.8%**	57.4%	87.4%	83.6%
GPT 5.6 Luna	86.5%	**89.0%**	86.4%	83.1%	84.8%	83.8%	86.6%	86.2%
Muse Spark 1.2	94.2%	**98.6%**	96.4%	97.2%	87.2%	66.2%	87.3%	79.9%
Qwen3.7 Flash	84.1%	68.8%	86.4%	80.2%	86.4%	29.3%	**94.9%**	78.8%

Highlighted cell is each model's best harness on its own balanced task set: Codex for DeepSeek V4, GPT 5.6 Luna, Muse Spark 1.2; OMP for GLM 5.2; Pi for Qwen3.7 Flash.

SECTION 03 — FINDINGS 02 AND 03

Rankings do not transfer

If harness quality were a property of the harness, every model would rank the eight the same way. It does not. How far from agreement depends on how you balance the task sets, so all three specifications are given.

+0.286	+0.336	+0.400
UNBALANCED	**PER-PAIR BALANCED**	**GLOBAL COMMON SET**
as published, all tasks	tasks both models ran	25 tasks, all 40 stacks

Mean Kendall τ between the harness rankings induced by each pair of models. The correct reading is the middle figure: +0.336 is weak positive agreement. Harness rankings carry some signal across models and nowhere near enough to transfer a recommendation. The unbalanced +0.286 overstates the effect and the 25-task +0.400 understates it by dropping the tasks that most separate harnesses.

Harness rank by model, balanced within each model

Harness	DeepSeek V4	GLM 5.2	GPT 5.6 Luna	Muse Spark 1.2	Qwen3.7 Flash	Range
Claude Code	6	2	3	4	4	**4**
Codex	1	6	1	1	7	**6**
Droid	3	4	4	3	2	**2**
Hermes	5	3	8	2	5	**6**
OMP	2	1	6	6	3	**5**
OpenCode	8	8	7	8	8	**1**
Pi	4	5	2	5	1	**4**
Prime Agent	7	7	5	7	6	**2**

> **THE ASYMMETRY**
>
> There is no universally good harness in this data, and one that is bad almost everywhere. OpenCode ranks last for 4 of the five models and seventh for the fifth, 20 points below the next harness on mean rate. Codex is first for DeepSeek V4, GPT 5.6 Luna, Muse Spark 1.2 yet sixth for GLM 5.2 and seventh for Qwen3.7. Hermes is second for Muse Spark and last for GPT 5.6. Excellence here is contextual; failure is nearly absolute.

The same non-transfer appears along a second axis. Ranking harnesses within each task family and comparing those rankings gives mean τ = +0.170. A harness that is strong at refactoring tells you little about whether it is strong at systems code.

SECTION 04 — FINDING 04

Determinism is the signal

Most cells ran three to five times. In 308 of 1230 multi-trial cells (25.0%) the same harness, on the same model, on the same task, both passed and failed. Correlating each harness's flake rate against its mean pass rate gives the strongest relationship in the dataset.

Harness	Non-deterministic cells	Flake rate	Mean strict rate
Droid	23 / 153	15.0%	**88.5%**
Pi	24 / 150	16.0%	**87.8%**
Codex	33 / 155	21.3%	**86.3%**
OMP	33 / 153	21.6%	**86.9%**
Claude Code	35 / 153	22.9%	**87.1%**
Hermes	41 / 155	26.5%	**86.3%**
Prime Agent	45 / 154	29.2%	**81.7%**
OpenCode	74 / 157	47.1%	**61.7%**

-0.950

ALL 8 HARNESSES

flake rate vs pass rate

-0.839

EXCLUDING OPENCODE

7 points, outlier removed

+0.315

TOKENS VS PASS RATE

the nearest rival predictor

TWO HONEST CAVEATS

First, the headline -0.950 is outlier-sensitive: remove OpenCode and seven points give -0.839. Still the strongest predictor measured, but not the near-perfect relationship the full-set figure implies. Second, the metric is partly mechanical: a cell at 0% or 100% cannot be flaky, so mid-range cells are structurally over-represented. The effect survives both, because the ordering holds among harnesses with near-identical means (Droid flakes on 15.0% of cells at 88.5%; Hermes on 26.5% at 86.3%).

The practical consequence stands regardless of mechanism. Single-run evaluation misses a quarter of the phenomenon. Five runs of one task is a cheaper and higher-signal screen than one run of thirty six.

SECTION 05 — FINDING 05

Spending does not buy correctness

Relationship	Pearson r	Reading
Judged quality × pass rate	+0.829	craft tracks correctness in aggregate
Tokens × pass rate	+0.315	verbosity buys little
Cost × pass rate	+0.219	price is weak signal
Wall-clock × pass rate	+0.115	latency is not capability

Cost per passing task spans 126× across paid stacks, from $0.0028 (Qwen3.7 Flash + Pi) to $0.3453 (GLM 5.2 + Codex), for the same problems. Within a single model, extra verbosity buys little and can be negative: GPT 5.6 on Codex burns 5.7× the tokens of GPT 5.6 on Pi for +2.3 points; GLM 5.2 on Codex burns 4.7× for -2.1 points.

Strictly dominated configurations

12 domination pairs across 5 distinct configurations. A configuration is dominated when another harness on the same model costs at least 3× less and also passes more. GLM 5.2 + Codex alone accounts for four of the pairs, which is why the pair count and the configuration count differ.

Model	Dominated	Cost / pass	Beaten by	Cost / pass	Cheaper
DeepSeek V4	Hermes	$0.0243 / 82.7%	OMP	$0.0076 / 85.3%	3.2×
DeepSeek V4	Hermes	$0.0243 / 82.7%	Pi	$0.0027 / 82.9%	8.8×
GLM 5.2	Codex	$0.2946 / 85.3%	Claude Code	$0.0568 / 88.7%	5.2×
GLM 5.2	Codex	$0.2946 / 85.3%	Pi	$0.0413 / 87.4%	7.1×
GPT 5.6 Luna	Hermes	$0.0787 / 83.1%	Claude Code	$0.0204 / 86.5%	3.9×
GPT 5.6 Luna	Hermes	$0.0787 / 83.1%	OpenCode	$0.0248 / 83.8%	3.2×
GPT 5.6 Luna	Hermes	$0.0787 / 83.1%	Pi	$0.0183 / 86.6%	4.3×
GPT 5.6 Luna	Hermes	$0.0787 / 83.1%	Prime Agent	$0.0202 / 86.2%	3.9×
GPT 5.6 Luna	OMP	$0.0713 / 84.8%	Claude Code	$0.0204 / 86.5%	3.5×
GPT 5.6 Luna	OMP	$0.0713 / 84.8%	Pi	$0.0183 / 86.6%	3.9×
GPT 5.6 Luna	OMP	$0.0713 / 84.8%	Prime Agent	$0.0202 / 86.2%	3.5×
Qwen3.7 Flash	Hermes	$0.0082 / 80.2%	Pi	$0.0026 / 94.9%	3.1×

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THE VALUE OUTLIER

The cheapest route above 90% in the matrix is Qwen3.7 Flash on Pi: $0.0026 per task at 94.9%. It beats every GPT 5.6 and GLM 5.2 configuration measured, while the strongest GPT 5.6 stack (Codex, 89.0%) costs 49× more per task and the strongest GLM 5.2 stack (OMP, 90.8%) costs 64× more. Muse Spark reaches 98.6% at zero billed cost, but it is a Contributor Edition model and is excluded from all price comparisons.

SECTION 06 — FINDING 06

Saturated suites stop measuring

Of 33 measured tasks, 10 sit at or above 95%. Discriminative power collapses as tasks saturate: the between-stack standard deviation falls from 0.258 on the hard band to 0.086 on saturated tasks, a factor of 3.0.

Task band	Tasks	Stacks	Mean	SD	Range
Saturated (≥95%)	10	40	96.7%	0.086	50.0% – 100.0%
Main suite, unsaturated	18	40	81.2%	0.152	17.3% – 97.4%
Hard band (33-37)	5	32	44.2%	0.258	**0.0% – 100.0%**

Rank the 32 common stacks on the main suite, rank them again on the five hard tasks, and compare: Kendall τ = +0.149. The two rankings are unrelated. An external audit confirmed this survives 2,000 random tie-break permutations.

The reshuffle

Stack	Main suite	Hard band	Move
GPT 5.6 Luna + Droid	#4 (96.0%)	#30 (6.7%)	-26
GPT 5.6 Luna + OMP	#8 (94.6%)	#32 (0.0%)	-24
GLM 5.2 + Pi	#7 (94.7%)	#26 (23.1%)	-19
GPT 5.6 Luna + Pi	#9 (94.5%)	#28 (20.0%)	-19
GPT 5.6 Luna + Hermes	#16 (91.7%)	#31 (6.7%)	-15
DeepSeek V4 + Hermes	#27 (86.7%)	#14 (46.7%)	+13
GPT 5.6 Luna + OpenCode	#25 (87.4%)	#10 (53.3%)	+15
Muse Spark 1.2 + OpenCode	#31 (65.4%)	#6 (73.3%)	+25

The judge fails where you need it

Every diff was scored 1-5 by an LLM judge alongside the oracle. Definitions matter here and earlier drafts blurred them. Using the published stack quality field across all tasks, r = +0.829. Using the mean of cell-level quality restricted to the main suite, r = +0.922. Using the same definition restricted to the hard band, r = +0.321.

THE WORST POSSIBLE FAILURE PROFILE

On the same aggregation, the judge scores +0.922 against ground truth on the main suite and +0.321 on the hard band. It is accurate on work easy enough that you could check it yourself and unreliable on work where a cheap proxy would be valuable. GPT 5.6 + Hermes earns 4.05 of 5 for code quality on the hard band while passing 1 attempt in 15. It is not fooled by bad code; it is fooled by good code that is wrong.

SECTION 07

What agents cannot do yet

Task family	Strict rate	Runs	Bar
systems	62.4%	752	
performance	73.4%	278	
parser	74.9%	279	
bugfix	75.0%	539	
refactor	79.8%	662	
feature	94.4%	1,804	
agentic-ops	95.0%	919	

A 33-point gap separates the work agents are demonstrated on from the work that holds a system together. By platform the pattern repeats: web APIs 97.8%, CLIs 93.6%, libraries 79.8%. Agents are strong at the edges of a system and weak in its core.

> **THE CEILING: 27_IR_OPTIMIZER**
>
> Zero passes across five models, eight harnesses and every trial. The task requires a compiler IR optimizer that simultaneously preserves exact trap semantics, remaps every absolute jump target under instruction removal, is idempotent, never grows the program, runs in near-linear time and does not mutate its input. Six coupled invariants, checked by a 469-line hidden oracle. Each is individually easy; holding all six at once is currently outside the frontier. A task with no variance cannot discriminate, so its cells are not in the shipped pack at all. On one pairing the container was OOM-killed at every memory tier from 1.5 GB to 8 GB with time-to-death flat across caps, so the failure mode is not a wrong answer but an inability to hold the problem.

Declared difficulty is unreliable

One task labelled mid before any run is harder than most tasks labelled hard: 06_async_bug at 30.2% sits below 9 of the 11 hard tasks. Human intuition about agent difficulty fails at exactly the resolution benchmark design requires.

Craft and correctness diverge by task type

Task	Pass rate	Judged quality	Gap
06_async_bug	30.2%	3.42 / 5	+0.38
03_rename_refactor	38.1%	3.25 / 5	+0.27
33_query_optimizer	34.0%	2.97 / 5	+0.25
15_race_condition	97.2%	3.61 / 5	-0.25
09_event_stream	96.6%	3.59 / 5	-0.25
08_safe_file_store	96.1%	3.56 / 5	-0.25

Async fixes and renames produce beautiful code that breaks. Concurrency and durability work produces unlovely code that holds. Any review process that reads the diff instead of running it inverts both populations.

What adversarial review changed

Two independent adversarial audits were run against the previous draft and its data pack. Both re-executed the pipeline, both recomputed the statistics with independent code, and both returned HOLD. Their quantitative verdict was that the arithmetic reproduces: one checked 122 figures and reproduced 118 exactly, with the remainder resolving in the pack's favour once definitions were mirrored. Their structural verdict was that the design disclosure and the integrity narrative did not hold up. A third launch-readiness audit then re-checked the measurement oracles themselves and removed four tasks from the export. Every finding is listed below with its resolution.

#	Finding	Resolution in this report
S1	Design described as fully crossed. It is not: only 15 tasks are common to all 40 stacks, and DeepSeek's Prime Agent ran 13 easy tasks its peers never saw.	Coverage disclosed on page 3 with a per-model table. All cross-model statistics now reported both unbalanced and balanced. DeepSeek Prime Agent's rank corrected from 2 to 7, and the coverage imbalance itself is repaired in this export.
S2	Pack promised zero-token cells are quarantined, yet ships 23 such cells (73 runs), two of which record passes.	Promise withdrawn and replaced with disclosure below. The two passing cells are flagged as suspect and their impact bounded.
S3	runs_sha256 matched no reconstructable artifact and no code computed it.	Replaced with a cell digest computed by the published script over the shipped cells, printed on the cover and reproducible in one command.
S4	Mean $\tau = +0.129$ is composition-sensitive; balanced sets give higher agreement.	Reported as a three-point band (+0.286 / +0.336 / +0.400) with the per-pair balanced figure named as the correct reading.
S5	'15 dominated stacks' were 15 pairs across 7 distinct configurations.	Restated as 12 pairs across 5 configurations, with the full pair list printed.
S6	Judge $r = +0.931$ used an aggregation different from the published stack quality field.	Both definitions now stated explicitly: +0.829 published field over all tasks, +0.922 cell-mean over the main suite.
S7	$r = -0.922$ collapses to -0.680 without the OpenCode outlier; not disclosed.	Both printed side by side on page 6.
S8	'288 cells of routing damage' reconciles with nothing in the shipped data.	Figure withdrawn. The verifiable scope is stated below.
S9	Unit errors throughout: runs described as cells, trials as cells, 495 excluded 'cells'.	Units corrected everywhere. The pack holds 1,247 cells covering 5,233 runs.
S10	'Moves the pass rate 28.5 points on average' was a mean best-to-worst range.	Restated with median (4.9pp) and mean (10.1pp) pairwise swap alongside the 31.1pp range.
S11	OOM exclusion row contradicted the shipped cells; Prime Agent × 29 is present and task 27 has no cells for any model.	Corrected below to the actual shortfall.
S12	Claim that only Qwen produced zero-token runs contradicted the shipped noop_runs column.	Withdrawn. Shipped counts printed below.

#	Finding	Resolution in this report
S13	Launch audit found three task verifiers unsound: 04's AST gate crashed on one harness's scaffolding (21 of 23 cells), and 11 and 12 passed an untouched starter file. The zero-token quarantine predicate also tracked the wrong signal.	Tasks 04, 11 and 12 removed from the export, and the zero-variance task 27 with them. The quarantine predicate was rebuilt with regression tests. The pack moved from 1,276 to 1,247 cells and every figure in this report was recomputed from the corrected data.

Integrity disclosures

Zero-token cells retained in the data (23 cells, 73 runs)

A cell recording zero tokens and zero cost cannot have consumed model capacity, so either usage accounting failed or the run did not happen as recorded. The previous draft claimed such cells are quarantined before publication. They are not. They are distributed across 4 models, so this is a platform-wide accounting gap rather than a single campaign's fault. Two record passes and are the only ones that can move a rate:

Model	Harness	Task	Result	Status
DeepSeek V4	Claude Code	28_lease_lock	2/5 pass	suspect
GLM 5.2	Claude Code	35_ts_to_python_port	3/3 pass	suspect

Bounding the impact: removing both cells changes the global strict rate by less than 0.1 points and changes no ranking in this report. The GLM + Claude case is the more serious of the two because it records 3 of 3 passes on a hard-band task, which is exactly the kind of observation the hard-band analysis rests on. GLM + Claude's hard-band rank would fall if that cell were dropped. Readers weighting the hard-band section should treat that single stack as unconfirmed.

Reported noop runs by model

The shipped data records agent runs that made no tool calls. The previous draft asserted these were Qwen-specific. The shipped column says otherwise: every model records them, and Qwen is not the extreme:

Model	DeepSeek V4	GLM 5.2	GPT 5.6 Luna	Muse Spark 1.2	Qwen3.7 Flash
noop runs	94	94	37	80	68

Routing defect: the verifiable scope

During the Qwen campaign, three harnesses returned near-zero pass rates at zero cost. The pinned endpoint was rejecting two request parameters, every call returned 404, the agent made no tool calls and the verifier honestly recorded failures. After the parameters were added to the proxy drop list and the affected cells re-measured, Qwen + Pi moved from a poisoned measurement to 94.9%, the cheapest configuration above 90% in the study. What is verifiable from the shipped pack: the three affected harnesses account for 80 shipped cells and 239 runs. The previously published figure of 288 affected cells is an operational count from campaign logs that cannot be reconstructed from these artifacts, and it is withdrawn.

> **THE LESSON THAT SURVIVES**
>
> A routing bug wearing a model-failure costume was concealing the best-value stack in the study, and it was caught only because a zero-dollar result was treated as a contradiction rather than a finding. That heuristic is sound and worth adopting. The irony, surfaced by adversarial review, is that the same pack then shipped 23 zero-token cells of its own. Nobody audits a bad result as hard as a good one, including the people who write that sentence.

Corrected exclusion record

Exclusion	Scope	Basis
27_ir_optimizer	all models, all harnesses	Zero passes anywhere. A task with no variance cannot discriminate; no cells are present in the pack.
03_rename_refactor	6 Qwen harnesses + affected cells elsewhere	Oracle imported the pre-rename symbol, so correct solutions failed. 21 valid runs survive.
29_incremental_parser × Pi	Qwen only, 1 cell	Container OOM reproduced at 1500m, 3g, 4g, 6g and 8g. Time-to-death flat across caps, indicating an allocation spike. Recorded as infrastructure failure, not model failure.

Qwen's grid is 7 cells short of its own 28- task universe: Codex on 03_rename_refactor, Droid on 03_rename_refactor, Hermes on 03_rename_refactor, OMP on 03_rename_refactor, Pi on 03_rename_refactor, Prime Agent on 03_rename_refactor, Pi on 29_incremental_parser. The previous draft described this as '7 cells on tasks 27 and 29 for Pi and Prime Agent', which the shipped data contradicts.

SECTION 09

Limits, and how to reproduce

Threats to validity

1. Five models is a small sample of model space. Every cross-model statistic rests on 5 points. The design is complete within each campaign and the effect sizes are large, but a sixth model could move these materially. Treat all cross-model correlations as descriptive, not inferential.
2. Task coverage is unbalanced. See page 3. Balanced recomputations are given for every affected statistic, and the balanced set is only 25 tasks.
3. Each harness ran in its default configuration, so part of the interaction term is 'this harness's default prompt suits this model'. That is a real deployment effect and not an intrinsic property of either component.
4. The hard band is 5 tasks across 4 models, with per-stack depth of 13 to 15 trials and 3 stacks below the full 15. τ over 32 stacks is the load-bearing number; individual hard-band rates are noisy.
5. The performance and parser families rest on 2 tasks each, roughly 28 to 42 runs per harness. Indicative only.
6. Cost is list-price derived, not invoiced. Ratios within a model are more robust than absolute dollars. Muse Spark has no billed cost and is excluded from all price statements.
7. The judge is a single model on a single rubric. A second judge would test whether the hard-band degradation is a property of LLM judging or of this judge.
8. Flake rate and pass rate are not fully independent by construction, as set out on page 6.
9. Qwen carries 645 runs against 1,113 to 1,186 for the others, so its intervals are wider.
10. 23 zero-token cells remain in the data, as disclosed on page 9.

Reproduction

The published pack contains the two source artifacts, an 18-file CSV pack including every cell, the generator that rebuilds all derived figures, and a claim verifier. Both scripts are deterministic and read only the two JSON inputs.

Command	Effect
python3 build_report_data.py	Rebuilds findings.json and all 18 CSVs from source
python3 verify_claims.py	Asserts every quoted figure against the data; non-zero exit on mismatch
python3 build_final_pdf.py	Regenerates this document, recomputing every number

Cell digest, sha256 over the 1,247 canonicalised cells covering 5,233 runs:

2bb0147855006219f8fa990a1cca81be9b442abc8761ca383c7f0730a9e29fcc

BurnBench

Which agent harness × model actually ships.

5,233 verified runs

1,247 cells · 40 stacks · 33 tasks